Approximation of Regular Functionals of Probability Measures by Interaction Polynomials

1 Functionals of measures and Wasserstein polynomials

Let $K \subset \mathbb{R}^d$ be compact and let

$$\mathcal{P}(K) = \{\mu : \mu \text{ is a Borel probability measure on } K\}.$$

Fix $q \geq 1$. We equip $\mathcal{P}(K)$ with the q-Wasserstein distance

$$W_q(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int_{K \times K} |x - y|^q \, d\pi(x, y) \right)^{1/q}.$$

We study the approximation of a functional

$$f : \mathcal{P}(K) \longrightarrow \mathbb{R}.$$

Unless stated otherwise, f is Borel measurable and bounded. For a bounded functional g, we use

$$\|g\|_\infty = \sup_{\mu \in \mathcal{P}(K)} |g(\mu)|.$$

1.1 Interaction polynomials

For $m \geq 1$ and a bounded measurable kernel

$$\Phi_m : K^m \longrightarrow \mathbb{R},$$

define

$$P_{\Phi_m}(\mu) = \int_{K^m} \Phi_m(x_1, \ldots, x_m) \, d\mu(x_1) \cdots d\mu(x_m).$$

Such a functional is called an interaction polynomial, or a Wasserstein polynomial, of order at most m.

Since $\mu^{\otimes m}$ is invariant under permutations, the kernel may always be symmetrized:

$$\Phi_m^{\mathrm{sym}}(x_1, \ldots, x_m) = \frac{1}{m!} \sum_{\sigma \in \mathfrak{S}_m} \Phi_m(x_{\sigma(1)}, \ldots, x_{\sigma(m)}).$$

The order m is the degree in the measure variable. It should not be confused with the ordinary polynomial degree of Φ_m in the spatial variables $x_1, \ldots, x_m$.

1.2 Empirical-measure polynomial

Given $x = (x_1, \dots, x_n) \in K^n$, set

$$\mu_x^n = \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}.$$

The canonical particle kernel associated with f is

$$\Phi_n^f(x_1, \dots, x_n) = f\left(\mu_x^n\right).$$

The corresponding particle polynomial is

$$B_n f(\mu) = \int_{K^n} f\left(\frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}\right) d\mu^{\otimes n}(x_1, \dots, x_n).$$

Equivalently, if $X_1, \dots, X_n$ are independent with law μ, then

$$B_n f(\mu) = \mathbb{E} f(\widehat{\mu}_n), \qquad \widehat{\mu}_n = \frac{1}{n} \sum_{i=1}^{n} \delta_{X_i}.$$

Thus $B_n f$ is a Wasserstein polynomial of order at most n.

This operator has two complementary classical interpretations. If K is finite, it is the multivariate Bernstein operator on the probability simplex. For general compact K, it is the canonical Bernstein–Schnabl operator on the Bauer simplex $\mathcal{P}(K)$: one samples the extreme points δ_{X_i} and evaluates f at their barycentric average. Probability-measure versions of this construction go back at least to Schnabl [10]; the uniform convergence of the empirical formulation for continuous f is also recorded in [1].

In statistics, $f(\widehat{\mu}_n)$ is the plug-in functional of the empirical measure. The weak expansion of its expectation and the higher-order particle correction used in Section 4.3 were developed systematically by Chassagneux, Szpruch, and Tse [4]. Their analysis uses high-order linear functional derivatives. Below we combine this weak-expansion mechanism with a separate Wasserstein control of the Taylor remainder.

2 Wasserstein–Hölder and Taylor regularity

2.1 Hölder regularity of order $0 < \alpha \leq 1$

Definition 1 (Metric Wasserstein–Hölder regularity). *For $0 < \alpha \leq 1$, the functional f is α-Hölder with respect to W_q if*

$$|f(\mu) - f(\nu)| \leq L W_q(\mu, \nu)^{\alpha}, \qquad \mu, \nu \in \mathcal{P}(K).$$

Proposition 1 (Hölder stability of the empirical operator). *If f is α-Hölder with constant L, then, for every $n \geq 1$ and $\mu \in \mathcal{P}(K)$,*

$$|B_n f(\mu) - f(\mu)| \leq L \mathbb{E} W_q(\widehat{\mu}_n, \mu)^{\alpha}.$$

Proof. Using $B_n f(\mu) = \mathbb{E} f(\widehat{\mu}_n)$, we obtain

$$\begin{aligned} |B_n f(\mu) - f(\mu)| &= \left|\mathbb{E}\left[f(\widehat{\mu}_n) - f(\mu)\right]\right| \\ &\leq \mathbb{E}\left|f(\widehat{\mu}_n) - f(\mu)\right| \\ &\leq L \mathbb{E} W_q(\widehat{\mu}_n, \mu)^{\alpha}. \end{aligned}$$

□

A literal metric Hölder condition with exponent larger than one is too restrictive on a geodesic metric space: subdividing a constant-speed geodesic into N pieces shows that the total oscillation is bounded by $LN^{1-\alpha}$ times the α-th power of its length, which tends to zero. In particular, this applies to $(\mathcal{P}(K), W_q)$ when the underlying space is geodesic, for example when K is convex. For a general compact K, the Wasserstein space need not be geodesic; in either case, higher differentiability is expressed by subtracting a Taylor polynomial rather than by imposing a metric exponent above one.

2.2 Wasserstein Taylor regularity of arbitrary order

For $\alpha > 1$, write

$$\alpha = r + \theta, \qquad r = \lceil \alpha \rceil - 1 \in \mathbb{N}, \qquad 0 < \theta \le 1.$$

Thus an integer $\alpha = m$ denotes the Lipschitz endpoint $C^{m-1,1}$.

Let $\mathcal{M}_0(K)$ be the vector space of finite signed Borel measures on K with total mass zero. For $\mu, \nu \in \mathcal{P}(K)$, write

$$\mu_t = (1-t)\mu + t\nu, \qquad 0 \le t \le 1.$$

Definition 2 (Flat/vertical linear functional derivatives)**.** *Set $D^0 f = f$. We say that f has linear functional derivatives up to order r if there are symmetric j-linear forms*

$$D^j f(\mu) : \mathcal{M}_0(K)^j \longrightarrow \mathbb{R}, \qquad 1 \le j \le r,$$

such that, for every mixture segment μ_t and every fixed $\eta_1, \dots, \eta_{j-1} \in \mathcal{M}_0(K)$,

$$\frac{d}{dt} D^{j-1} f(\mu_t)[\eta_1, \dots, \eta_{j-1}] = D^j f(\mu_t)[\nu - \mu, \eta_1, \dots, \eta_{j-1}].$$

For $j = 1$, the empty argument list on $D^0 f = f$ is omitted. Derivatives at $t = 0, 1$ are understood one-sided. Thus $D^j f(\mu)[\eta_1, \dots, \eta_j]$ is the mixed directional derivative whenever the perturbations are admissible; the multilinear form is assumed to extend to all of $\mathcal{M}_0(K)^j$. Whenever continuity or an operator norm is invoked, the norm on $\mathcal{M}_0(K)$ will be stated explicitly.

These derivatives act along affine mixture paths: they change the mass assigned to fixed points rather than transporting the points. They are therefore iterated *flat* or *vertical* derivatives. More precisely, at first order the definition above is the Gâteaux-like flat derivative on $\mathcal{P}(K)$. A stronger Fréchet vertical derivative also requires a uniform first-order remainder in a specified norm, such as total variation; that stronger property is not implicit in the definition above. We follow the terminology and distinctions summarized in [2].

When the derivatives admit kernels, we write

$$D^j f(\mu)[\eta_1, \dots, \eta_j] = \int_{K^j} \frac{\delta^j f}{\delta \mu^j}(\mu, x_1, \dots, x_j)\, d\eta_1(x_1) \cdots d\eta_j(x_j).$$

The kernels are also called linear functional or von Mises derivatives; this statistical functional calculus goes back to [12] and is treated systematically in [11]. Since every η_i has zero mass, the kernel is not unique. We choose the centred representative

$$\left(\frac{\delta^j f}{\delta \mu^j}\right)_c (\mu, x_1, \dots, x_j) = D^j f(\mu)[\delta_{x_1} - \mu, \dots, \delta_{x_j} - \mu].$$

It represents the same multilinear form and satisfies

$$\int_K \left(\frac{\delta^j f}{\delta \mu^j}\right)_c (\mu, x_1, \dots, x_j)\, d\mu(x_i) = 0$$

for each i, with the remaining variables held fixed. We omit the subscript c for this representative below.

There is an important distinction from the horizontal, or Lions, derivative. If $K \subset \mathbb{R}^d$ and the first vertical kernel is sufficiently regular in x, then

$$D_\mu^{\mathrm{H}} f(\mu, x) = \nabla_x \frac{\delta f}{\delta \mu}(\mu, x).$$

The vector-valued derivative on the left tests transport perturbations and, under the standard lift to random variables, agrees with the Lions derivative. Thus the scalar vertical derivative and the Lions derivative are not two names for the same object: the latter is obtained from the former by a spatial gradient when the required regularity is available. See [2, 3]. All Taylor and particle expansions in this note use the vertical derivatives.

The Taylor polynomial of order r around μ is

$$T_\mu^r f(\nu) = f(\mu) + \sum_{j=1}^{r} \frac{1}{j!} D^j f(\mu) \left[(\nu - \mu)^{\otimes j}\right].$$

Definition 3 (Wasserstein–Taylor class). *We say that $f \in C_W^\alpha$ if its Taylor polynomial of order r satisfies*

$$\left|f(\nu) - T_\mu^r f(\nu)\right| \leq L W_q(\mu, \nu)^\alpha$$

uniformly over $\mu, \nu \in \mathcal{P}(K)$.

The remainder estimate is the definition; an operator norm must be specified before stating a derivative-level criterion. On $\mathcal{M}_0(K)$, define the Kantorovich–Rubinstein norm

$$\|\eta\|_{\mathrm{KR}} = \sup_{\mathrm{Lip}(\varphi) \leq 1} \left| \int_K \varphi \, d\eta \right|.$$

In particular,

$$\|\nu - \mu\|_{\mathrm{KR}} = W_1(\mu, \nu) \leq W_q(\mu, \nu).$$

For an r-linear form A, use the induced operator norm

$$\|A\|_{\mathrm{op,KR}} = \sup_{\|\eta_i\|_{\mathrm{KR}} \leq 1} \left|A[\eta_1, \ldots, \eta_r]\right|.$$

This is not the $L^\infty(K^r)$ norm of a derivative kernel: a kernel supremum norm controls total-variation increments and does not supply the r powers of W_q needed below.

A sufficient condition for membership in C_W^α is

$$\|D^r f(\mu) - D^r f(\nu)\|_{\mathrm{op,KR}} \leq L W_q(\mu, \nu)^\theta.$$

Indeed, with $\Delta = \nu - \mu$, the integral Taylor formula gives

$$f(\nu) - T_\mu^r f(\nu) = \frac{1}{(r-1)!} \int_0^1 (1-t)^{r-1} \big(D^r f(\mu_t) - D^r f(\mu)\big)[\Delta^{\otimes r}] \, dt.$$

Using the operator norm,

$$|f(\nu) - T_\mu^r f(\nu)| \leq \frac{L}{(r-1)!} \int_0^1 (1-t)^{r-1} W_q(\mu_t, \mu)^\theta \, \|\Delta\|_{\mathrm{KR}}^r \, dt.$$

The coupling that leaves a mass $1 - t$ fixed and transports the remaining mass gives

$$W_q(\mu_t, \mu) \leq t^{1/q} W_q(\mu, \nu).$$

Consequently,

$$|f(\nu) - T_\mu^r f(\nu)| \le C_{r,\theta,q} L W_q(\mu,\nu)^\theta W_1(\mu,\nu)^r \le C_{r,\theta,q} L W_q(\mu,\nu)^{r+\theta}.$$

The converse need not hold without additional extension and differentiability assumptions, which is why this criterion is not stated as an equivalence.

This is the analogue on $\mathcal{P}(K)$ of the classical decomposition

$$\alpha = r + \theta$$

for functions in a Hölder space $C^{r,\theta}$.

Remark 1 (Relation with $C^{1,\theta}$ terminology). *When $r = 1$, the fractional exponent in the derivative-level condition is θ, and the total Taylor order is $\alpha = 1 + \theta$. Thus the classes denoted $C^{1,\theta}$ in [2] have total order $1 + \theta$ in the convention of this note. They should not, however, be identified automatically with $C_W^{1+\theta}$: Bertucci's vertical class uses total variation in the measure variable and the supremum norm of the flat kernel, while his horizontal class controls the Lions derivative along couplings. Our sufficient criterion instead combines a vertical derivative, the Kantorovich–Rubinstein operator norm, and a W_q^θ modulus.*

2.3 Taylor regularity in a root-n tangent norm

The functional derivatives are the same linear derivatives as above. What changes is the geometry used to control the remainder. We keep r for the Taylor degree throughout, rather than introducing a second symbol.

Fix an integer $r \ge 1$. Suppose that

$$f(\nu) = f(\mu) + \sum_{j=1}^{r} \frac{1}{j!} D^j f(\mu)\left[(\nu-\mu)^{\otimes j}\right] + R_{r+1}(\mu,\nu),$$

and that centred derivative kernels can be chosen so that

$$\sup_{\mu\in\mathcal{P}(K)} \left\| \frac{\delta^j f}{\delta\mu^j}(\mu,\cdot) \right\|_{L^\infty(K^j)} < \infty, \qquad 1 \le j \le r.$$

This boundedness is used later to control the expected Taylor contractions; the root-n remainder estimate itself uses only the next two assumptions.

Let $\|\cdot\|_{\mathcal{T}}$ be a norm or seminorm on $\mathcal{M}_0(K)$ for which the empirical fluctuation has root-n moments:

$$\sup_{\mu\in\mathcal{P}(K)} \mathbb{E}\,\|\widehat{\mu}_n - \mu\|_{\mathcal{T}}^{r+1} \le C_r n^{-(r+1)/2}.$$

Assume that

$$|R_{r+1}(\mu,\nu)| \le C\,\|\nu-\mu\|_{\mathcal{T}}^{r+1}.$$

Then

$$\sup_{\mu\in\mathcal{P}(K)} \mathbb{E}|R_{r+1}(\mu,\widehat{\mu}_n)| \le C C_r n^{-(r+1)/2}.$$

The comparison with a Wasserstein remainder requires an explicit relation between the two geometries. In the typical situation

$$\|\nu-\mu\|_{\mathcal{T}} \le C_{\mathcal{T}} W_q(\mu,\nu),$$

the tangent norm is weaker, so the upper bound $C\,\|\nu-\mu\|_{\mathcal{T}}^{r+1}$ is smaller and therefore more demanding than a W_q^{r+1} remainder bound. The tangent estimate implies the Wasserstein one, but the converse generally fails. Along empirical measures, the tangent hypothesis gives the dimension-free scale $n^{-(r+1)/2}$, whereas a bare Wasserstein-remainder hypothesis gives $\rho_{n,q,d}^{r+1}$, defined in Section 4.1 below.

For example, given finitely many bounded Lipschitz functions $\varphi_1, \dots, \varphi_L$, the finite-moment seminorm

$$\|\eta\|_\Phi = \left(\sum_{\ell=1}^{L} \left| \int_K \varphi_\ell \, d\eta \right|^2 \right)^{1/2}$$

has root-n empirical moments and satisfies $\|\nu - \mu\|_\Phi \lesssim W_1(\mu, \nu) \le W_q(\mu, \nu)$. It may vanish for two distinct measures having the same selected moments. Requiring the entire Taylor remainder to be controlled by this much smaller quantity is therefore a genuine structural restriction on f, not merely a smoother choice of notation.

3 Examples of functionals

3.1 Linear moments

For a bounded function $V : K \to \mathbb{R}$, let

$$f(\mu) = \int_K V(x) \, d\mu(x).$$

This is a polynomial of order one. Moreover,

$$B_n f(\mu) = f(\mu)$$

for every n, so the approximation error is zero.

3.2 Pair interactions

Let $W : K \times K \to \mathbb{R}$ be bounded and Borel measurable, and define

$$f(\mu) = \iint_{K \times K} W(x, y) \, d\mu(x) d\mu(y).$$

This is a polynomial of order two. The particle approximation satisfies the exact identity

$$B_n f(\mu) - f(\mu) = \frac{1}{n} \left[\int_K W(x, x) \, d\mu(x) - \iint_{K \times K} W(x, y) \, d\mu(x) d\mu(y) \right].$$

The diagonal bias can be removed by the U-statistic

$$\frac{1}{n(n-1)} \sum_{i \ne j} W(X_i, X_j).$$

This is the classical passage from a V-statistic to a U-statistic; see [7, 11].

3.3 Smooth functions of finitely many moments

Let $\varphi_1, \dots, \varphi_L : K \to \mathbb{R}$ be bounded and Lipschitz, and let $F : \mathbb{R}^L \to \mathbb{R}$. Consider

$$f(\mu) = F\left(\int_K \varphi_1 \, d\mu, \dots, \int_K \varphi_L \, d\mu \right).$$

If $F \in C^{r,\theta}$ on a neighborhood of the compact set of attainable moment vectors m_μ, with bounded derivatives there, then f has functional derivatives of the same order. For example,

$$\frac{\delta f}{\delta \mu}(\mu, x) = \sum_{\ell=1}^{L} \partial_\ell F(m_\mu) \left(\varphi_\ell(x) - \int_K \varphi_\ell \, d\mu \right),$$

where

$$m_\mu = \left(\int \varphi_1 \, d\mu, \ldots, \int \varphi_L \, d\mu \right).$$

These are prototypical high-regularity functionals for which empirical fluctuations are finite dimensional and therefore of order $n^{-1/2}$, independently of d.

3.4 Squared maximum mean discrepancy

Let k be a bounded positive-definite kernel with reproducing kernel Hilbert space H, and define

$$m_\mu = \int_K k(x, \cdot) \, d\mu(x).$$

For a fixed ν, let

$$f_\nu(\mu) = \mathrm{MMD}_k^2(\mu, \nu) = \|m_\mu - m_\nu\|_H^2 .$$

Expanding the square gives

$$f_\nu(\mu) = \iint k(x, y) \, d\mu(x) d\mu(y) - 2 \iint k(x, y) \, d\mu(x) d\nu(y) + C_\nu.$$

Thus squared MMD is exactly a polynomial of order two.

Its particle bias is

$$B_n f_\nu(\mu) - f_\nu(\mu) = \frac{1}{n} \left[\int k(x, x) \, d\mu(x) - \iint k(x, y) \, d\mu(x) d\mu(y) \right].$$

The corresponding U-statistic removes this bias exactly.

The unsquared MMD,

$$\mathrm{MMD}_k(\mu, \nu) = \|m_\mu - m_\nu\|_H ,$$

is not differentiable at $\mu = \nu$. At the null,

$$\mathbb{E} \, \mathrm{MMD}_k(\widehat{\mu}_n, \mu) = O(n^{-1/2}),$$

whereas away from the null a Taylor expansion can give a weak bias of order $1/n$.

3.5 Squared Wasserstein distance

Let

$$f_\nu(\mu) = W_2^2(\mu, \nu).$$

If $D = \mathrm{diam}(K)$, then

$$\left| W_2^2(\mu, \nu) - W_2^2(\mu', \nu) \right| \leq 2D W_2(\mu, \mu').$$

Therefore

$$|B_n f_\nu(\mu) - f_\nu(\mu)| \leq 2D \mathbb{E} W_2(\widehat{\mu}_n, \mu).$$

The functional $W_2^2(\cdot, \nu)$ is not globally smooth in the von Mises sense. Its differentiability depends on uniqueness and regularity of the optimal transport map.

At $\nu = \mu$,

$$B_n f_\mu(\mu) - f_\mu(\mu) = \mathbb{E}W_2^2(\widehat{\mu}_n, \mu).$$

Uniformly over compactly supported measures,

$$\mathbb{E}W_2^2(\widehat{\mu}_n, \mu) \lesssim \begin{cases} n^{-1/2}, & d < 4, \\ n^{-1/2}\log(1+n), & d = 4, \\ n^{-2/d}, & d > 4. \end{cases}$$

These uniform compact-support bounds follow, for example, from the empirical Wasserstein estimates in [5, 6].

There are special linear cases. For example,

$$W_2^2(\mu, \delta_a) = \int_K |x-a|^2 \, d\mu(x),$$

and consequently the particle approximation is exact.

3.6 Relative entropy

This example is a warning outside the standing bounded real-valued setting: without restrictions or regularization, relative entropy is extended valued.

For

$$f_\nu(\mu) = \mathrm{KL}(\mu \| \nu),$$

the direct empirical kernel is generally not finite. If ν is nonatomic and has a density, then

$$\widehat{\mu}_n \not\ll \nu$$

and hence

$$\mathrm{KL}(\widehat{\mu}_n \| \nu) = +\infty$$

almost surely.

On a finite state space with S states,

$$f_{p^\star}(p) = \sum_{j=1}^{S} p_j \log \frac{p_j}{p_j^\star}$$

is smooth in the interior of the simplex. If all probabilities are bounded away from zero, then

$$\mathbb{E}f_{p^\star}(\widehat{p}_n) - f_{p^\star}(p) = \frac{S-1}{2n} + O(n^{-2}).$$

For continuous measures, one may instead regularize:

$$f_{\nu,\varepsilon}(\mu) = \mathrm{KL}(K_\varepsilon \mu \| \nu),$$

where $K_\varepsilon \mu$ is a smoothed density. For fixed $\varepsilon > 0$, this functional can possess bounded functional derivatives and a $1/n$ weak expansion. The constants generally grow as a power of ε^{-d}.

4 Rate of the particle interaction polynomial

4.1 Empirical Wasserstein scale and the raw particle error

Define the characteristic empirical Wasserstein scale

$$\rho_{n,q,d} = \begin{cases} n^{-1/(2q)}, & d < 2q, \\ n^{-1/(2q)}(\log(1+n))^{1/q}, & d = 2q, \\ n^{-1/d}, & d > 2q. \end{cases}$$

For every fixed $a > 0$, compact support gives the uniform moment estimate

$$\sup_{\mu \in \mathcal{P}(K)} \left(\mathbb{E}W_q(\widehat{\mu}_n, \mu)^a\right)^{1/a} \leq C_{a,q,d,K}\rho_{n,q,d};$$

this is a uniform ambient-dimensional upper bound [5, 6]. It need not be sharp when K is finite or has smaller intrinsic dimension.

The smooth weak-bias component of the result below is the static i.i.d. empirical-measure expansion developed in [4]. The proposition combines that cancellation mechanism with the Wasserstein remainder, thereby also covering fractional and dimension-dependent regularity regimes.

Proposition 2 (Raw particle error across regularity regimes)**.** *Let $\alpha > 0$. Assume the metric Hölder condition of Section 2.1 when $0 < \alpha \leq 1$. When $\alpha > 1$, assume $f \in C^\alpha_W$, with $r = \lceil \alpha \rceil - 1$, and assume that each $D^j f$, $1 \leq j \leq r$, admits a centred kernel that is uniformly bounded over $\mu \in \mathcal{P}(K)$. Then*

$$\boxed{\|B_n f - f\|_\infty \lesssim \begin{cases} \rho^\alpha_{n,q,d}, & 0 < \alpha \leq 2, \\ n^{-1} + \rho^\alpha_{n,q,d}, & \alpha > 2. \end{cases}}$$

Proof. For $0 < \alpha \leq 1$, Proposition 1 and the empirical moment estimate give the result. Let $\alpha > 1$, put

$$\Delta_n = \widehat{\mu}_n - \mu, \qquad Z_i = \delta_{X_i} - \mu,$$

and use the Taylor expansion

$$f(\widehat{\mu}_n) - f(\mu) = \sum_{j=1}^{r} \frac{1}{j!} D^j f(\mu)[\Delta_n^{\otimes j}] + R_\alpha(\mu, \widehat{\mu}_n).$$

The first-order term is centred:

$$\mathbb{E}Df(\mu)[\Delta_n] = 0.$$

For $j \geq 2$, multilinearity gives

$$\mathbb{E}D^j f(\mu)[\Delta_n^{\otimes j}] = \frac{1}{n^j} \sum_{i_1,\ldots,i_j=1}^{n} \mathbb{E}D^j f(\mu)[Z_{i_1}, \ldots, Z_{i_j}].$$

A summand vanishes if one particle index occurs exactly once, by independence and $\mathbb{E}Z_i = 0$. A surviving tuple therefore contains at most $\lfloor j/2 \rfloor$ distinct indices. Uniform boundedness of the derivative kernel then yields

$$\left|\mathbb{E}D^j f(\mu)[\Delta_n^{\otimes j}]\right| \lesssim n^{-j+\lfloor j/2 \rfloor} = n^{-\lceil j/2 \rceil}.$$

Finally,

$$\mathbb{E}|R_\alpha(\mu, \widehat{\mu}_n)| \lesssim \rho^\alpha_{n,q,d}.$$

For $1 < \alpha \leq 2$, one has $r = 1$, so only the centred linear term appears. For $\alpha > 2$, all terms of order $j \geq 2$ are $O(n^{-1})$. $\square$

In the high-dimensional regime $d > 2q$, the proposition becomes

$$\boxed{\|B_n f - f\|_\infty \lesssim \begin{cases} n^{-\alpha/d}, & 0 < \alpha \le 2, \\ n^{-\min\{1,\alpha/d\}}, & \alpha > 2. \end{cases}}$$

Thus Section 4.1 is not restricted to $\alpha \le 1$: the first line uses metric Hölder regularity for $0 < \alpha \le 1$ and the first-order Taylor remainder for $1 < \alpha \le 2$. The second line uses a genuine second derivative and higher-order Taylor regularity.

4.2 The quadratic bias and the $1/n$ threshold

When $D^2 f$ exists, the quadratic contraction can be written exactly as

$$\frac{1}{2}\mathbb{E}D^2 f(\mu)[\Delta_n^{\otimes 2}] = \frac{A_1 f(\mu)}{n},$$

where

$$A_1 f(\mu) = \frac{1}{2}\int_K D^2 f(\mu)[\delta_x - \mu, \delta_x - \mu]\, d\mu(x).$$

Under sufficiently high linear, or von Mises, Taylor regularity with a remainder smaller than the retained terms, the remaining contractions have an expansion in integer powers of $1/n$, beginning with

$$B_n f(\mu) - f(\mu) = \frac{A_1 f(\mu)}{n} + \frac{A_2 f(\mu)}{n^2} + \cdots .$$

This is the classical von Mises expansion of a plug-in functional [12, 11]. An all-order treatment of this static empirical-measure expansion under high-order linear functional derivatives, together with particle applications, is given in [4].

This explains the saturation statement precisely. If the terms beyond $A_1 f/n$ are $o(n^{-1})$ and $A_1 f$ is not identically zero, the uncorrected operator is generically of exact order $1/n$; additional derivatives do not cancel $A_1 f$. For example, for

$$f(\mu) = \left(\int_K \varphi\, d\mu\right)^2,$$

one has the exact identity

$$B_n f(\mu) - f(\mu) = \frac{\mathrm{Var}_\mu(\varphi)}{n},$$

although f is already a polynomial and has derivatives of every order. Linear functionals and other special cases may have $A_1 f = 0$, so $1/n$ is not a universal lower bound.

Under Wasserstein–Taylor regularity alone, the remainder $\rho^\alpha_{n,q,d}$ may be larger than $1/n$. Thus $\alpha = 2$ by itself does not imply a $1/n$ rate. In the regime $d > 2q$, the Wasserstein remainder is $O(n^{-1})$ only when $\alpha \ge d$. For $d < 2q$, the corresponding condition is $\alpha \ge 2q$; at the critical dimension $d = 2q$, equality still carries a logarithmic loss. By contrast, the root-n tangent regularity of Section 2.3 gives a dimension-free $O(n^{-1})$ raw bound already from a quadratic tangent remainder.

4.3 Higher-order corrected construction

Let $J \ge 2$ denote the desired *weak correction order*; it is not the smoothness index α. Suppose that $r = \lceil \alpha \rceil - 1 \ge 2J - 2$, the derivative kernels through order r are uniformly bounded, and the C^α_W remainder bound holds. The same index-partition argument as above groups the Taylor contractions into

$$B_m f = f + \sum_{\ell=1}^{J-1} \frac{A_\ell f}{m^\ell} + \mathcal{E}_{m,J},$$

with

$$\|\mathcal{E}_{m,J}\|_\infty \le C\left(m^{-J} + \rho^\alpha_{m,q,d}\right).$$

Indeed, the coefficient $A_\ell f$ involves derivatives only through order 2ℓ; terms whose first possible weak order is at least J enter $\mathcal{E}_{m,J}$. The strict condition $\alpha > 2J-2$ is an equivalent convenient way to write $r \ge 2J-2$ under the convention of Section 2.2.

Choose distinct positive integers

$$\lambda_1, \dots, \lambda_J$$

and coefficients $a_1, \dots, a_J$ satisfying

$$\sum_{j=1}^J a_j = 1, \qquad \sum_{j=1}^J a_j \lambda_j^{-\ell} = 0, \quad 1 \le \ell \le J-1.$$

This Vandermonde system has a unique solution. Define the Richardson–Romberg particle polynomial

$$\mathcal{B}_{n,J} f = \sum_{j=1}^J a_j B_{\lambda_j n} f.$$

This is Richardson extrapolation [8] applied to the weak particle expansion; its use in this measure-functional setting follows [4]. It is a signed interaction polynomial of order at most

$$\max_j \lambda_j n.$$

For instance, $J=2$, $(\lambda_1, \lambda_2) = (1,2)$, gives

$$\mathcal{B}_{n,2} f = 2B_{2n} f - B_n f,$$

which cancels the $A_1 f/n$ term.

In general, the first $J-1$ weak bias terms cancel, and

$$\boxed{\|\mathcal{B}_{n,J} f - f\|_\infty \lesssim n^{-J} + \rho^\alpha_{n,q,d}.}$$

In the high-dimensional regime $d > 2q$,

$$\boxed{\|\mathcal{B}_{n,J} f - f\|_\infty \lesssim n^{-\min\{J, \alpha/d\}}.}$$

The minimum has three distinct meanings:

- If $\alpha/d \le 1$, the Wasserstein remainder term already controls the general raw-operator bound, so correction does not improve the guaranteed exponent.
- If $1 < \alpha/d < J$, correction removes the $1/n$ saturation and the bound reaches the full $n^{-\alpha/d}$ Wasserstein remainder rate.
- If $\alpha/d \ge J$, the corrected weak-bias term n^{-J} controls the displayed bound; increasing J improves the guarantee if the required derivatives are available.

Thus, when $J \geq \alpha/d$, the minimum is indeed α/d, but this means that the construction has successfully removed the weak bias term from this upper bound and reached the general Wasserstein remainder ceiling. It does not mean that the correction failed to use smoothness. For example, if $\alpha/d = 2.4$, the guaranteed raw exponent is 1, correction order $J = 2$ gives exponent 2, and $J = 3$ reaches exponent 2.4. Special functionals with vanishing contractions or a sharper remainder can, of course, converge faster than these general bounds.

Remark 2 (Relation with the Remez algorithm)**.** *The correction above is neither the Remez algorithm nor a direct extension of it. Classical Remez approximation fixes a polynomial degree p and computes a minimax approximant*

$$q_p^\star \in \arg\min_{q \in \Pi_p} \|g - q\|_{L^\infty([a,b])},$$

where Π_p is the space of univariate polynomials of degree at most p. Its exchange iteration enforces the Chebyshev equioscillation condition at $p+2$ adaptively chosen extremal points [9]. By contrast, the coefficients a_j in $\mathcal{B}_{n,J}$ are fixed by cancelling the first $J-1$ powers in the weak particle-bias expansion. There is no exchange of extremal measures, no equioscillation condition, and no claim that $\mathcal{B}_{n,J} f$ is a best uniform approximation among interaction polynomials of a prescribed order. The limited analogy is that both methods determine signed coefficients through a structured linear system in order to reduce a uniform approximation error.

A genuine Remez-type problem on $\mathcal{P}(K)$ would first choose a finite-dimensional space $\mathcal{V}_N$ of interaction or moment polynomials and solve

$$P_N^\star \in \arg\min_{P \in \mathcal{V}_N} \|f - P\|_\infty .$$

That is a different, generally infinite-dimensional minimax problem unless $\mathcal{V}_N$ is fixed explicitly. Remez is more directly relevant to the second approximation level of Section 5: for a fixed particle order n, it may be used to compute a minimax spatial kernel $Q_{n,p}$ when that kernel depends on a single scalar variable, possibly after a structural reduction. For a generic kernel on $K^n \subset \mathbb{R}^{nd}$, the classical one-dimensional equioscillation theorem does not apply directly. Thus the particle/Richardson stage exploits smoothness in the measure variable, whereas a Remez step could optimize the separate spatial-kernel approximation.

4.4 Dimension-free weak Taylor rate

Suppose instead that the degree-r Taylor expansion satisfies the root-n tangent assumptions of Section 2.3. Put

$$\sigma_r = \frac{r+1}{2}, \qquad J_r = \lceil \sigma_r \rceil.$$

Each expected Taylor contraction is a finite sum of integer powers of $1/n$, and its smallest possible power is $\lceil j/2 \rceil$ at derivative order j. Consequently,

$$B_m f = f + \sum_{1 \leq \ell < \sigma_r} \frac{A_\ell f}{m^\ell} + \mathcal{R}_{m,r}, \qquad \|\mathcal{R}_{m,r}\|_\infty \lesssim m^{-\sigma_r}.$$

Using J_r Richardson levels and cancelling the powers $\ell = 1, \dots, J_r - 1$ therefore gives, for the previously fixed levels λ_j, the corrected operator $\mathcal{B}_{n,J_r}$, with the convention $\mathcal{B}_{n,1} = B_n$, and the dimension-free bound

$$\boxed{\|\mathcal{B}_{n,J_r} f - f\|_\infty \lesssim n^{-(r+1)/2}.}$$

There is no need to replace this by a floor exponent: the possible half-integer rate comes from the explicitly assumed $(r+1)$-st moment of the tangent remainder. Roughly two additional bounded functional derivatives are needed for each additional power of $1/n$. Without correction, the generic rate is $1/n$: for $r \geq 2$ the quadratic contraction remains, while for $r = 1$ the assumed quadratic tangent remainder itself is $O(n^{-1})$.

5 Two-level approximation of the interaction kernel

Let $P_n f$ denote a chosen first-level interaction polynomial of order at most n. For $B_n f$, this is its natural particle order. For a corrected operator $\mathcal{B}_{m,J}$, set

$$\Lambda_J = \max_{1\le j\le J} \lambda_j, \qquad n = \Lambda_J m,$$

and lift every lower-order term by adding dummy variables; the result is then $P_n f = \mathcal{B}_{m,J} f$. Since J and the levels λ_j are fixed, $n \asymp m$, so this reindexing does not change any algebraic rate. Write the resulting kernel as

$$P_n f(\mu) = \int_{K^n} \Phi_n(x_1,\dots,x_n)\, d\mu^{\otimes n}.$$

Although each particle belongs to $\mathbb{R}^d$, the complete kernel is a function on

$$K^n \subset \mathbb{R}^{nd}.$$

Approximating a generic kernel is therefore an nd-dimensional approximation problem.

5.1 Ordinary polynomial approximation of the kernel

Let $Q_{n,p}$ be an ordinary polynomial in the nd scalar variables, using a specified tensor-degree or total-degree convention with degree parameter p. Define

$$P_{n,p} f(\mu) = \int_{K^n} Q_{n,p}(x_1,\dots,x_n)\, d\mu^{\otimes n}.$$

Assume that Φ_n has spatial smoothness $s > 0$ and that, for some $\gamma \ge 0$,

$$\boxed{\|\Phi_n - Q_{n,p}\|_\infty \le C n^\gamma p^{-s}.}$$

The factor n^γ allows the spatial smoothness norm of the kernel to grow with the number of particles.

Measure-variable regularity and spatial regularity are distinct: bounded functional derivatives of f do not automatically imply uniform C^s-regularity of Φ_n as a function on K^n.

Suppose the first approximation level satisfies

$$\|f - P_n f\|_\infty \le C n^{-a}.$$

Then

$$\begin{aligned}\|f - P_{n,p} f\|_\infty &\le \|f - P_n f\|_\infty + \|\Phi_n - Q_{n,p}\|_\infty \\ &\le C_0 n^{-a} + C_1 n^\gamma p^{-s}.\end{aligned}$$

Consequently,

$$\boxed{\|f - P_{n,p} f\|_\infty \le C_0 n^{-a} + C_1 n^\gamma p^{-s}.}$$

The relevant first-level exponent is, for example,

$$a = \begin{cases} \alpha/d, & \text{for the unsaturated high-dimensional rate, } 0<\alpha\le 2,\\ \min\{1,\alpha/d\}, & \text{for the uncorrected high-dimensional rate when } \alpha>2,\\ \min\{J,\alpha/d\}, & \text{for weak correction order } J\text{, assuming } \alpha > 2J-2,\\ 1, & \text{for an uncorrected root-}n\text{ tangent remainder},\\ (r+1)/2, & \text{for a corrected root-}n\text{ Taylor expansion of degree } r. \end{cases}$$

In the first row, regularity means metric Hölder regularity when $0<\alpha\le 1$ and Wasserstein–Taylor regularity when $1<\alpha\le 2$.

5.2 Optimal choice of p for a fixed n

Balancing the two errors,

$$n^{-a} \asymp n^{\gamma} p^{-s},$$

gives

$$\boxed{p \asymp n^{(a+\gamma)/s}.}$$

With this choice,

$$\boxed{\|f - P_{n,p} f\|_{\infty} \lesssim n^{-a}.}$$

In the stable case $\gamma = 0$,

$$p \asymp n^{a/s}.$$

For an error tolerance ε,

$$n \asymp \varepsilon^{-1/a}, \qquad p \asymp \varepsilon^{-(a+\gamma)/(as)}.$$

When $\gamma = 0$,

$$p \asymp \varepsilon^{-1/s}.$$

For example, suppose the first-level construction reaches the high-dimensional Wasserstein remainder exponent α/d, as it does for $0 < \alpha \leq 2$, or after correction when $J \geq \alpha/d$ under the hypotheses of Section 4.3. In the stable case $\gamma = 0$, if

$$a = \frac{\alpha}{d} \qquad \text{and} \qquad s = \alpha,$$

then

$$\boxed{p \asymp n^{1/d}, \qquad \|f - P_{n,p} f\|_{\infty} \lesssim n^{-\alpha/d}.}$$

Likewise, if $\gamma = 0$ and the corrected tangent Taylor rate is

$$a = \frac{r+1}{2}$$

and the spatial smoothness is $s = r + 1$, then

$$\boxed{p \asymp n^{1/2}, \qquad \|f - P_{n,p} f\|_{\infty} \lesssim n^{-(r+1)/2}.}$$

5.3 Representation in terms of ordinary moments

Write the polynomial kernel as

$$Q_{n,p}(x_1, \ldots, x_n) = \sum_{\ell=1}^{M} c_{\ell} \prod_{i=1}^{n} x_i^{a_{\ell,i}},$$

where

$$a_{\ell,i} \in \mathbb{N}^d, \qquad x_i^{a_{\ell,i}} = \prod_{j=1}^{d} x_{i,j}^{a_{\ell,i,j}}.$$

Integration against $\mu^{\otimes n}$ gives

$$P_{n,p} f(\mu) = \sum_{\ell=1}^{M} c_{\ell} \prod_{i=1}^{n} m_{a_{\ell,i}}(\mu),$$

where

$$m_a(\mu) = \int_K x^a \, d\mu(x)$$

is an ordinary moment of μ.

Thus the two-level construction eventually produces an ordinary polynomial in finitely many moments.

5.4 Number of coefficients

A full tensor polynomial with degree at most p in each of the nd scalar variables has

$$\boxed{M_{\text{tensor}}(n,p) = (p+1)^{nd}}$$

coefficients.

A polynomial of total degree at most p has

$$\boxed{M_{\text{total}}(n,p) = \binom{nd+p}{p}.}$$

For a generic s-smooth function of nd variables, the classical fixed-dimensional approximation relation can be expressed as

$$\text{kernel error} \lesssim M^{-s/(nd)}.$$

Using this estimate while n varies is an additional uniformity assumption: all constants depending on the growing dimension nd must be controlled by the displayed factor n^γ. Such control does not follow from the fixed-dimensional theorem alone.

Let $P_{n,M}f$ denote the resulting two-level approximation when its ordinary polynomial kernel has M coefficients. A parameter-count version of the estimate is

$$\boxed{\|f - P_{n,M}f\|_\infty \lesssim n^{-a} + n^\gamma M^{-s/(nd)}.}$$

This formula displays the principal difficulty: the exponent $s/(nd)$ deteriorates as the particle order n increases.

5.5 Optimal rate as a function of the total parameter count

Balancing

$$n^{-a} \asymp n^\gamma M^{-s/(nd)}$$

gives

$$(a+\gamma)\log n \asymp \frac{s}{nd}\log M.$$

Equivalently,

$$n \log n \asymp \frac{s}{d(a+\gamma)}\log M.$$

Hence

$$n \asymp \frac{\log M}{\log\log M},$$

up to constants depending on a, s, d, γ. Substitution into the first-level error gives

$$\boxed{\|f - P_{n,M}f\|_\infty \lesssim \left(\frac{\log M}{\log\log M}\right)^{-a}.}$$

If the first-level construction reaches the high-dimensional Wasserstein remainder rate—for instance, when $0 < \alpha \le 2$, or after correction with $J \ge \alpha/d$ under the hypotheses of Section 4.3—then

$$a = \frac{\alpha}{d},$$

this becomes

$$\boxed{\|f - P_{n,M}f\|_\infty \lesssim \left(\frac{\log M}{\log\log M}\right)^{-\alpha/d}.}$$

For a corrected Taylor approximation with

$$a = \frac{r+1}{2},$$

one obtains

$$\boxed{\|f - P_{n,M}f\|_\infty \lesssim \left(\frac{\log M}{\log \log M}\right)^{-(r+1)/2}.}$$

When the hypotheses of Section 4.1 or Section 2.3 genuinely give the uncorrected exponent $a = 1$, one has

$$\boxed{\|f - P_{n,M}f\|_\infty \lesssim \frac{\log \log M}{\log M}.}$$

6 Conclusion

There are three different sources of approximation error.

1. The empirical-measure error:

   $$W_q(\widehat{\mu}_n, \mu),$$

   which makes a metric Hölder remainder or a C_W^α Taylor remainder contribute the high-dimensional rate

   $$n^{-\alpha/d}.$$

   For the raw operator this term must be combined with the weak bias, as in Section 4.1.

2. The weak particle bias:

   $$\frac{A_1 f}{n} + \frac{A_2 f}{n^2} + \cdots,$$

   which causes a generic uncorrected approximation to saturate at $1/n$ once a weak expansion makes the remaining error $o(n^{-1})$. Higher-order corrections remove these contractions and are required to exploit additional Taylor regularity beyond this threshold.

3. The approximation of the kernel Φ_n, which is a function of nd scalar variables. Under the full tensor convention, the number of coefficients grows exponentially with nd at fixed per-coordinate degree; other conventions still face a deteriorating dimension-dependent approximation exponent.

As a function of the interaction order n, one can obtain algebraic rates such as

$$n^{-\alpha/d} \qquad \text{or} \qquad n^{-(r+1)/2}.$$

Under the generic kernel-approximation model of Section 5, the construction gives the logarithmic parameter-count upper bound

$$\left(\frac{\log M}{\log \log M}\right)^{-a}.$$

To obtain an algebraic rate in M by this route, one needs additional kernel structure, such as symmetry reduction, sparse interactions, dependence on finitely many moments, low-rank tensor decompositions, or kernels that are already finite-order polynomials, such as squared MMD and pair-interaction energies.

References

[1] G. J. Babu, *A note on Dubins' theorem*, Canadian Mathematical Bulletin **33**(4) (1990), 416–418.

[2] C. Bertucci, *Analysis on spaces of measures*, arXiv:2607.19939, 2026.

[3] R. Carmona and F. Delarue, *Probabilistic Theory of Mean Field Games with Applications I: Mean Field FBSDEs, Control, and Games*, Probability Theory and Stochastic Modelling, vol. 83, Springer, 2018.

[4] J.-F. Chassagneux, Ł. Szpruch, and A. Tse, *Weak quantitative propagation of chaos via differential calculus on the space of measures*, The Annals of Applied Probability **32**(3) (2022), 1929–1969.

[5] S. Dereich, M. Scheutzow, and R. Schottstedt, *Constructive quantization: approximation by empirical measures*, Annales de l'Institut Henri Poincaré, Probabilités et Statistiques **49**(4) (2013), 1183–1203.

[6] N. Fournier and A. Guillin, *On the rate of convergence in Wasserstein distance of the empirical measure*, Probability Theory and Related Fields **162** (2015), 707–738.

[7] W. Hoeffding, *A class of statistics with asymptotically normal distribution*, The Annals of Mathematical Statistics **19**(3) (1948), 293–325.

[8] L. F. Richardson, *The approximate arithmetical solution by finite differences of physical problems involving differential equations, with an application to the stresses in a masonry dam*, Philosophical Transactions of the Royal Society of London, Series A **210** (1911), 307–357.

[9] E. Ya. Remez, *Sur la détermination des polynômes d'approximation de degré donnée*, Communications de la Société Mathématique de Kharkov **10** (1934), 41–63.

[10] R. Schnabl, *Zur Approximation durch Bernsteinpolynome auf gewissen Räumen von Wahrscheinlichkeitsmaßen*, Mathematische Annalen **180** (1969), 326–330.

[11] R. J. Serfling, *Approximation Theorems of Mathematical Statistics*, Wiley, 1980.

[12] R. von Mises, *On the asymptotic distribution of differentiable statistical functions*, The Annals of Mathematical Statistics **18**(3) (1947), 309–348.